Tuesday Reading Assessment: Chapters 6, 7

Prof. Jordan C. Hanson

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1 Logic Functions: Multiplexers

1. Draw a diagram of the gate logic describing the 4-to-1 multiplexer, using basic AND, OR, and inverter gates. The design should include the correct number of data-select lines, four inputs, and one output.

2. Suppose that the data lines are called D_i, the data select lines are S_i, and the output line is D_{out}. If the S_i represent binary counting between 0-3, with 1 μs per clock step, how often can one send a bit of data to D_{out} using the D_0 line?

3. How long would it take to send four bits of data, one on each D_i?

2 Logic Functions: Demultiplexers

1. Draw a diagram of the gate logic describing the 1-to-4 demultiplexer, using basic AND, OR, and inverter gates. The design should include the correct number of data-select lines, one input, and four outputs.

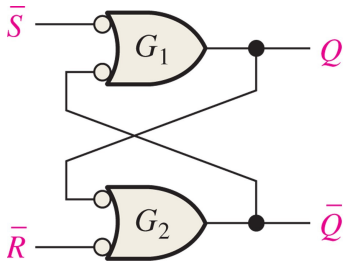


Figure 1: A schematic of an SR-latch.

2. Suppose that the data input line is called D_{in}, the data output lines are D_i the data select lines are S_i. The S_i represent binary counting between 0-3, with 1 μs per clock step. Create a timing diagram showing the behavior of D_{in}, S_i, and D_i.

3 Latches, Flip-Flops, and Timers

1. Consider 1, in which the basic schematic of an SR-latch is shown. Suppose, initially, $Q = 0$, and the wire labeled $\bar{R}$ is HIGH. If $\bar{S}$ momentarily goes LOW, what will happen to Q? To $\bar{Q}$?